

Task 1: List the IP address classes (A, B, C, D, E) and give one real-world example of a device or service that might use each class. Write your answers in a table format.

Class	Address range	Default subnet mask	Real-world example
A	1.0.0.0 – 126.255.255.255	255.0.0.0 (/8)	Large-scale backbone networks needing millions of addresses, like an ISP's core infrastructure
B	128.0.0.0 – 191.255.255.255	255.255.0.0 (/16)	A large organization's network, like a university campus connecting thousands of devices
C	192.0.0.0 – 223.255.255.255	255.255.255.0 (/24)	Your home Jio Fiber router's network — most home and small-office networks (192.168.x.x) use this class
D	224.0.0.0 – 239.255.255.255	Not used for hosts	Multicast traffic
E	240.0.0.0 – 255.255.255.255	Not used for hosts	Reserved for experimental/research use — not assigned to real devices

Task 2: Given the IP address 192.168.10.45 with subnet mask 255.255.255.0, calculate the network address, broadcast address, and the range of valid host addresses. Hint: Show your working steps for each calculation.

IP Add: 192.168.10.45

Subnet Mask: 255.255.255.0

This is a Class C - Private address

Subnet mask in Binary: 11111111.11111111.11111111.00000000

N=0 , H=8

No. of Subnet = $2^N = 1$

No. of Hosts: $2^H = 256$

IP Address Range:

192.168.10.0	Network Address
192.168.10.1	Usable Host Address
to	
192.168.10.254	
192.168.10.255	Broadcast Address

Task 3: Create a subnetting plan for a small food delivery startup like Zomato that needs 4 separate subnets: one for Delivery Partners, one for Restaurants, one for Customers, and one for Admins. Assign a unique subnet (with network address and subnet mask) for each group using the 10.0.0.0/24 network.

IP Address: 10.0.0.0/24

/24 - CIDR

Subnet: 255.255.255.0

No. of Network bits: 24

No. of Host bits: 8

Now to create 4 Subnets

No of Network Bits: $N=2$

Then No. of Host bits reduces: $H=6$

No. of Subnets: $2^N = 2^2 = 4$

No. of Hosts: $2^H = 2^6 = 64$ hosts

No. of Usable Hosts: $64 - 2 = 62$

Group	Network Address	First Usable IP	Last Usable IP	Broadcast Address	Full Range
Admins	10.0.0.0	10.0.0.1	10.0.0.62	10.0.0.63	10.0.0.0 - 10.0.0.63

Delivery Partners	10.0.0.64	10.0.0.65	10.0.0.126	10.0.0.127	10.0.0.64 - 10.0.0.127
Restaurants	10.0.0.128	10.0.0.129	10.0.0.190	10.0.0.191	10.0.0.128 - 10.0.0.191
Customers	10.0.0.192	10.0.0.193	10.0.0.254	10.0.0.255	10.0.0.192 - 10.0.0.255

Task 4 : You are given the following scenario: A college WiFi network uses the IP range 172.16.0.0/16. The admin wants to create at least 20 subnets for different buildings. Calculate the new subnet mask and the number of hosts available per subnet. Constraint: Show your calculation steps and final answers.

/16 - CIDR = No. of Network Bits

Subnet mask: 255.255.0.0

No. of Host bits: 16

Now to create 20 subnets:

No of Network bits is to be increased

Borrowing 5 Network bits from Host bits

No. of Subnets: $2^N = 2^5 = 32$

Host bits reduced from 16 to 11

No of Hosts = $2^H = 2^{11} = 2048$

New subnet mask: 11111111.11111111.11110000.000000

255.255.248.0

New CIDR - /21

Task 5: A friend sends you a message asking why their device with IP 192.168.1.300 is not connecting to the network. Write a short explanation identifying the error and what a valid IP address in that subnet should look like.

The error is that 300 is not a valid octet value. Every part of an IPv4 address must be between 0 and 255 (since each octet is 8 bits, max value $2^8 - 1 = 255$). 300 exceeds that limit.

A valid IP address in that subnet (assuming 192.168.1.0/24) would be anything from 192.168.1.1 to 192.168.1.254